Generation of Gaussian Quantum Steering and Entanglement in Coupled Lossy Waveguides

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(Spring 2024–2026)

Dedication

I dedicate this work to my parents, whose unconditional support and encouragement have been the backbone of my academic journey. I also dedicate it to my husband, whose confidence in me and constant support have greatly encouraged and strengthened my academic journey.

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Dr. Shakir Ullah, for his guidance, valuable suggestions, and constant support throughout this research. His knowledge in quantum optics and quantum information has helped me learn and think more deeply, and I am truly thankful for that. I am grateful to the Department of Physics, Quaid-e-Azam University, for providing a supportive research environment throughout this work. I would like to express my heartfelt gratitude to my parents, whose unwavering support and encouragement have been the backbone of my academic journey. A special thanks to my husband, whose patience, support, and motivation kept me going. His belief in me made this work possible. Lastly, I acknowledge the efforts of researchers whose previous studies laid the groundwork on which this research is built.

Abstract

Quantum correlations play an important role in quantum information science and are basis for the operation of modern quantum technologies. Particularly, quantum entanglement and quantum steering are key resources for continuous-variable quantum information processing. In this thesis, the covariance matrix formalism is used to study the time-dependent dynamics of quantum entanglement and Gaussian quantum steering in a coupled lossy optical waveguide system. The paper is concerned with realistic integrated photonic structures, taking the optical losses and environmental interactions into account.

The physical system consists of two optical waveguides which are coupled evanescently, the coherent energy exchange between the modes is described by the Hamiltonian while the dissipation and attenuation mechanisms are modeled by a Lindblad master equation that allows the system to be analyzed as an open quantum system. Within the Gaussian-state formalism, the first and the second moments of field quadratures give the full description of all the physical properties of the system. Quantum entanglement is quantified using logarithmic negativity (E_N), while quantum steering is quantified using the Gaussian steering measure (S).

It has been found that both quantum entanglement and quantum steering are oscillatory functions of propagation time due to the coherent transfer of energy between the coupled waveguide modes. These quantum correlations are however affected by dissipation and are found to be dependent on the loss rate (κ), the purity of the initial Gaussian state (μ), and the nonclassicality depth (τ). It is also found that quantum steering is more fragile than quantum entanglement in the same physical conditions, which proves the hierarchical nature of quantum correlations. In addition, increasing the coupling strength (J) is shown to partially compensate the degrading effects of losses.

These findings show that, despite realistic loss mechanisms, coupled optical waveguides can preserve nonclassical correlations over relevant propagation lengths. The results highlight the potential of integrated photonic platforms for applications in photonic quantum information processing, one-sided device-independent quantum key distribution, and quantum teleportation

TABLE OF CONTENTS

Acknowledgments viii

Abstract viii

List of Figures iv

List of Abbreviations v

1 Introduction 1

1.1 Background and Motivation . . . 2

1.2 Thesis Layout . . . 5

2 Literature Review 7

2.1 Quantum States Of Light . . . 8

2.1.1 Continuous Variable States . . . 8

2.1.2 Gaussian States . . . 8

2.1.3 Squeezed States . . . 9

2.1.3.1 Two-Mode Squeezing and Operator Correlations . . . 11

2.1.3.2 Connection to covariance matrices 11
2.2 Covariance Matrix 12
2.3 Optical Waveguides as Quantum Systems 13
2.3.1 Evanescent Overlap/Coupled Mode Theory 13
2.3.2 Loss and Decoherence in Real Waveguide Systems 14
2.3.3 Relevance Of Optical Waveguides For Quantum Circuits/Photonics 14
2.4 Quantum Steering 15
2.4.1 Quantification of Quantum Steering using Gaussian Steering Measure 16
2.5 Quantum Entanglement In Continuous Variable Systems 16
2.5.1 Entanglement Quantification Measures 17
2.6 Loss and Decoherence 18
2.6.1 Loss (Dissipation) and Pure Decoherence 18
2.6.2 Loss Mechanism in Waveguides 19
2.7 Applications and Motivations 19
2.7.1 Applications of Quantum Entanglement 20
2.7.2 Applications of Quantum Steering 20
2.7.3 Applications of Waveguides 21

3 Model System and Theoretical Formulations 23
3.1 The Model System 24
3.2 Hamiltonian of the Coupled Waveguide System 24
3.2.1 System Hamiltonian 24
3.2.2 Physical Interpretation of the Coupling 25
3.3 Open-System Dynamics and Loss Modelling 25
3.3.1 Quantum Liouville (Master) Equation 26
3.4 Dynamical Equations of the Model System 26
3.5 Analytical Solutions of the Moment Equations 27
3.6 Covariance Matrix and Quantification of Quantum Steering and Entanglement 29
3.6.0.1 Covariance Matrix of the Two-Mode System 30
3.7 Gaussian Quantum Steering 30
3.8 Gaussian Quantum Entanglement 31
3.9 Relation Between Entanglement and Steering 31

4 Results and Discussion 33

4.1 Time Evolution of Gaussian Quantum Steering and Entanglement . . . 34

4.2 Effect of Nonclassicality τ . . . 34

4.3 Effect of Purity μ . . . 36

4.4 Effect of Loss Parameter κ . . . 37

4.5 Gaussian Quantum Steering and Entanglement in Squeezed States . . . 38

4.6 Effect of coupling strength on Quantum Steering and Entanglement . . . 39

4.7 Summary . . . 41

5 Conclusion 43

6 References 45

List of Figures

3.1 Schematic diagram of a coupled waveguide system 24

4.1 Time evolution of Gaussian quantum steering and entanglement 34

4.2 Effect of nonclassicality τ on quantum correlations 35

4.3 Effect of purity μ on quantum correlations 36

4.4 Effect of loss parameter κ on quantum correlations 37

4.5 Quantum correlations for squeezed states (lossless) 38

4.6 Effect of loss on quantum correlations in squeezed states 39

4.7 Effect of coupling strength J on quantum correlations 40

List of Abbreviations

CV	Continuous Variable
DV	Discrete Variable
QIT	Quantum Information Theory
QKD	Quantum Key Distribution
1sDI	One-Sided Device-Independent
1sDI-QKD	One-Sided Device-Independent Quantum Key Distribution
EPR	Einstein–Podolsky–Rosen
LHS	Local Hidden State
PPT	Positive Partial Transpose
TMGS	Two-Mode Gaussian State
NOON	N-Photon Entangled State
HOM	Hong–Ou–Mandel Interference
E_N	Logarithmic Negativity (Entanglement Measure)
$S_{A\to B}$	Gaussian Quantum Steering from Party A to Party B
H_2	Rényi-2 Entropy
K	Secure Key Rate
ρ	Density Matrix
σ	Covariance Matrix
$\tilde{\sigma}$	Partially Transposed Covariance Matrix
Ω	Symplectic Matrix
ν	Symplectic Eigenvalue
$\tilde{\nu}$	Smallest Symplectic Eigenvalue
$\hat{a}, \hat{a}^\dagger$	Bosonic Annihilation and Creation Operators
Q, R	Field Quadrature Operators
$W(z)$	Wigner Function
J	Evanescent Coupling Strength Between Waveguides
κ	Optical Loss Rate (Damping Constant)

κ/J	Loss-to-Coupling Ratio
ω	Angular Frequency of Waveguide Mode
n_1, n_2	Mean Photon Numbers
m_1, m_2	Single-Mode Squeezing Moments
$c(t)$	Inter-Mode Coherence
$s(t)$	Two-Mode Correlation Function
τ	Nonclassical Depth
μ	Purity of the Gaussian State
r	Squeezing Parameter
θ	Squeezing Phase
MIQS	Measurement-Induced Quantum Steering
CNOT	Controlled-NOT Gate

Chapter 1

Introduction

1.1 Background and Motivation

Quantum correlations is one of the basic aspect of quantum mechanics required for many current technologies in quantum information [1, 12]. The field of quantum information started with the aim of controlling individual quantum bits (qubits), and has led to important concepts, such as quantum computing [1], quantum cryptography [2], and quantum teleportation [3].Later researchers discovered an immensely powerful alternative by using continuous-variable (CV) quantum information carriers instead of qubits, offering the quantization of degrees of freedom of bosonic systems, such as the electromagnetic field [11, 4]. These systems are crucial in quantum communication, as well as for very advanced methods, that are limited only by quantum physics, and are used to detect, sense, and image [11].

In quantum processes involving CVs, it is frequently of interest to consider the Gaussian realm, i.e., Gaussian states and Gaussian transformations, and this is a space where the mathematics is simple and well-organized. Further,the required optical devices in the laboratory,(such as beam splitters, phase shifters, and squeezers) are simple to access and are readily available in large quantities [11]. This area is referred to as Gaussian quantum information processing, and allows a vast range of important tasks to be performed. The most important of these protocols are quantum correlations, most notably entanglement and quantum steering, which are key to almost all quantum information protocols being implemented successfully [11].

In 1935 the idea of quantum entanglement gained much attention with the Einstein-Podolsky-Rosen (EPR) paradox raising the question of whether quantum mechanics provides a complete description of reality [7]. Later that year, Schrodinger referred to this effect as entanglement and also proposed a new form of quantum correlation which is now called quantum steering [8, 19]. These discoveries formed the basis of the modern theory of quantum correlation, which are crucial for various quantum information protocols [25, 1].

The phenomenon known as quantum steering was introduced in 1935 [8]. It is a form of quantum correlation that has received considerable attention in quantum information theory, especially after its formalization in 2007 [38]. Schrodinger first described this phenomenon as "magic" as the effect does not facilitate transmission of information, and it is the ability of one party (Alice) to measure or steer the wave function of another distant party [19]. One common notion in the modern theory is the inability of the conditional quantum states of one party (Bob) to be described by a local hidden state (LHS) model. A joint bipartite quantum state ϱ_{AB} is said to be steerable if Bob is forced to believe that Alice is influencing his system through some form of action at a distance. Conversely, when (LHS) exists for the shared state,then such a bipartite system is referred to as an unsteerable state [19].

One-sided device-independent quantum key distribution (QKD) is essential due to quantum steering [41] , in which one party is confident in their measurement equipment, whereas the other is not, which is clear in the possibility of secret key distillation and demonstrates the state steerability [41, 19]. It also has randomness certification in this asymmetric case with more extractable randomness than the Bell case with full device-independence [42]. Besides, steering offers an operational benefit in subchannel discrimination tasks and is conceptually equivalent to quantum teleportation and analysis of security in secret sharing protocols [44, 19].

Quantum entanglement is recognized as a fundamental essence of quantum mechanics and quantum information theory [25]. Entanglement is described as a physical phenomenon, in which the associated particles remain correlated irrespective of their physical distance between them [7]. This property is deemed important for the practical implementation of the quantum information processing systems. Its importance can be seen in numerous important technological domains such as,quantum computing, quantum cryptography, and quantum teleportation playing a significant part[1, 25]. Particularly, entanglement is viewed as being crucial to illustrate the supremacy of a quantum computer when comparing it to its classical analogs [45]. In connection to this,in coupled lossy waveguides, entanglement is an important concept, as these waveguides can be used as basic parts of quantum circuits [46, 47, 51]. Although the real devices always have some losses, the devices are still useful to quantum information tasks and it is therefore necessary to know how quantum entanglement and quantum steering really behaves in them [56, 28, 25].

An optical waveguide is a structure that guides light along a desired path.Coupled waveguides are very important because they act as basic building blocks for quantum circuits. These waveguides allow efficient control of light and can be effectively used to manipulate light to accomplish quantum information processing, where the manipulation of individual photons in a waveguide is fundamentally involved [46, 50, 47]. In addition, coupled silica waveguides have been extended to make multimode integrated interferometers which can produce arbitrary quantum circuits, as well as can produce two and four photon entangled states analogous to NOON states [51, 46, 52]. In addition to computation, such structures can be considered to study other important quantum effects such as quantum interference, entanglement and quantum walk [47, 53, 50]. An important point of these structures is that it is dealing with real material loss and it still finds that the waveguides hold up well. They are able to maintain their entanglement even though the loss is quite strong [28]. This shows that coupled lossy waveguides can still work as useful parts of quantum circuits, making them suitable for many quantum information applications [28, 11].

As quantum technologies have advanced, optical waveguide-based photonic platforms have become the focus of modern quantum hardware. In these waveguide based devices light is trapped in small, stable cores with high index, and small and extremely scalable quantum circuits are constructed [46, 50]. By making two waveguides close to each other, all the evanescent fields overlaps, leading to evanescent coupling mechanism that enables controlled energy exchange just like a beam splitter. This interaction is the foundation on which to apply on-chip quantum gates and other necessary photonic operations are applied [28, 46, 50].

however, in real devices, some amount of optical loss and noise from the environment is always present [56]. The energy of the photons decreases as they move, leading to an effective interaction with the empty vacuum modes; the interaction tends to destroy the quantum state of a system, and destroys the distinct quantum correlations between the modes [54, 56].Such perturbations destroy quantum correlations, i.e quantum entanglement and quantum steering [25, 19], and they pose an enormous limitation to photonic architectures which require either large propagation ranges or very precise interactions between gates [28, 11, 17].

For this reason, understanding how quantum correlations behave in coupled waveguides when they experience losses is essential for determining whether integrated quantum photonic systems can work in practice. By studying how entanglement and steering affects under an attenuation, we can evaluate how robust these quantum resources are and assess the viability of waveguide-based circuits for future quantum technologies.

This research analyzes the time-dependent dynamics of quantum steering and entanglement in a passive two-mode directional coupler with realistic loss, using the Gaussian-states and Gaussian transformation formalism. The study examines how several key parameters—including the loss coefficient κ, the initial nonclassicality τ, the purity

μ,coupling parameter J and the input squeezing r of the Gaussian state—collectively determine the preservation or degradation of quantum steering and entanglement as the fields propagate through lossy waveguides.

1.2 Thesis Layout

This thesis is structured as follows:

Chapter 2 begins with an overview of quantum states of light discussing CV quantum states particularly Gaussian states together with their representation through quadrature operators, wigner functions and covariance matrices. After that the discussion moves towards optical waveguides which highlights evanescent coupling , loss and decoherence in waveguides and applications of optical waveguides in quantum photonics. Moreover, the chapter then reviews the two important quantum correlations, i.e., quantum steering and quantum entanglement with their quantification,significance, and applications.Last but not the least, the applications and relevance of the optical waveguides in quantum circuits and photonics, and the applications of quantum steering and quantum entanglement are overviewed. After that Chapter 3 starts with an overview of our physical system.It then describes the systems Hamiltonian. It is followed by the discussion of the physical interpretation of the coupling. The chapter then discusses the open system dynamics and loss modelling which contains the quantum liouville equation and lindblad operator for waveguide loss.It then describes equations of motion for our system,the expectation value formalism and the moment equations and their solutions. With these moment equations in hands, we calculated covariance matrix and then quantified quantum steering and quantum entanglement. The quantum hierarchy relationship between entanglement and steering is also explained which is then proved from our results.A special type of Gaussian states called squeezed states, Its covariance matrix which we used to evaluate entanglement and steering is discussed at the end of this chapter. Chapter 4 is all about the results, begins the discussion about the dynamical behaviour of Gaussian quantum steering and entanglement in a coupled lossy waveguide system. First of all it took general case in which all the parameters κ, τ, μ,are kept constant. It then takes specific cases one by one to see the effect of τ,μ and κ where J=1 for all. This chapter then explains the results for the special type of Gaussian states such as squeezed states. Initially, we consider describes ideal non lossy case and then discusses the effect of losses on quantum steering and entanglement in squeezed states.

Lastly, the effect of coupling parameter J on quantum steering and quantum entanglement is explained.The chapter ends with the summary of the results and discussion. In Chapter 5 we have written the conclusion, where we discussed the full summary of the thesis research work and our key findings. All bibliography and citations are provided in Chapter 6.

Chapter 2

Literature Review

This chapter gives a review of the theoretical background of quantum correlations in integrated photonic circuits. It begins with a review of continuous-variable quantum states, particularly of Gaussian states, the properties of which are fully characterized by a covariance matrix, extracted out of the field quadrature. This is in turn followed by a discussion of the physical platform of optical waveguides, how the light is localized within the guiding medium and interaction between adjacent modes through evanescent interaction, provide ordinary optical attenuation (α) couples the system to the vacuum and causes the system to decay (Γ), making it hard to preserve quantum information. The chapter also summarizes two important quantum correlations investigated in this work, that is, Gaussian quantum steering, denoted by ($\mathcal{S}$), and quantum entanglement, characterized by the logarithmic negativity E_N.All the above factors together are the conceptual framework that are used in the dynamical development of these resources in coupled waveguide architectures. This chapter ends with the discussion of the vital uses of entanglement, steering, and waveguides.

2.1 Quantum States Of Light

Quantum states of light represent the quantized description of the electromagnetic field, where the field is treated as a bosonic system governed by quantum mechanics. These states are typically expressed using state vectors or density matrices [55]. These states, unlike classical light, can have nonclassical characteristics, including squeezing, entanglement and quantum correlations, which are of paramount importance to Continuous variable quantum information and quantum technologies [11]. Quantum states of light are classified into pure and mixed states [55, 11]. In terms of information encoding and processing quantum states of light can either be modeled in a discrete variable model, in terms of photon-number observable, or in a continuous variable model, where information is encoded in field quadrature with continuous spectra [4, 11].

2.1.1 Continuous Variable States

In quantum mechanics, a continuous variable state is a quantum state where the physical observable, i.e, momentum and position vary continuously unlike discrete variable state where observable can take only discrete values. These system are illustrated with in an infinite-dimensional Hilbert space [4]. In these system the principal variables are the quadrature operators which obey specific commutation relations that form the foundation of the system's quantum structure [10]. Continuous variable quantum states are classified into two categories: Gaussian and non-Gaussian states [12].

2.1.2 Gaussian States

In a Continuous variable framework, a Gaussian state is a special kind of quantum state whose Fourier representation has a Gaussian form in phase space. These states are completely described by their first and second moments [11]. The quadrature operators can be defined in terms of bosonic operators [55].

$$\hat{Q} = \frac{1}{\sqrt{2}}(\hat{a} + \hat{a}^\dagger), \qquad \hat{P} = \frac{i}{\sqrt{2}}(\hat{a}^\dagger - \hat{a}), \tag{2.1}$$

which satisfy the canonical commutation relation

$$[\hat{Q}, \hat{P}] = \frac{i}{\hbar}. \tag{2.2}$$

The Wigner function of a Gaussian state is

$$W(\hat{\mathbf{z}}) = \frac{1}{(2\pi)^m \sqrt{\det \boldsymbol{\sigma}}} \exp\left[-\frac{1}{2}(\hat{\mathbf{z}} - \bar{\mathbf{z}})^T \boldsymbol{\sigma}^{-1} (\hat{\mathbf{z}} - \bar{\mathbf{z}})\right]. \tag{2.3}$$

Here, m denotes the number of bosonic modes. The single-mode Wigner function is obtained by setting $m = 1$ in the above expression[10, 11].

and,

$$\hat{\mathbf{z}} = (\hat{Q}_1, \hat{P}_1, \dots, \hat{Q}_m, \hat{P}_m)^T$$

is the phase-space vector, whose components are denoted by $\hat{z}_i$,

$$\bar{\mathbf{z}} = \langle \hat{\mathbf{z}} \rangle. \tag{2.4}$$

is the displacement vector, and σ_{ij} is the co variance matrix, with matrix elements

$$\sigma_{ij} = \frac{1}{2}\langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle. \tag{2.5}$$

is the covariance matrix.

All physical properties of the Gaussian states, such as purity, entropy, steering and entanglement can be determined directly from the covariance matrix without full density operator because of the Gaussian form of its Wigner function. Typical examples of gaussian states include the coherent states, squeezed states, thermal states and vacuum state [11, 12].

2.1.3 Squeezed States

A special type of Gaussian states are squeezed states. As a Gaussian states, they are fully determined by their first and second statistical moments [11, 12]. The first moments are associated with the averages which are displacements operators in the phase space [11]. The second moments are grouped into a positive-definite, symmetric and real matrix called the covariance matrix [11, 12]. The action of a displacement and a squeezing operator on the vacuum state could create a squeezed state [12].

The theoretical description of squeezed states begins with the bosonic annihilation and creation operators $\hat{a}$ and $\hat{a}^\dagger$, which satisfy the canonical commutation relation

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

which leads to the uncertainty relation[39, 12]

$$\Delta Q\, \Delta P \geq \tfrac{1}{2}.$$

A single-mode squeezed vacuum state is generated by applying the unitary squeezing operator $\hat{S}(\zeta)$ to the vacuum state $|0\rangle$. The squeezing operator is defined as

$$\hat{S}(\zeta) = \exp\left[\tfrac{1}{2}\left(\zeta^* \hat{a}^2 - \zeta\, \hat{a}^{\dagger 2}\right)\right]. \tag{2.6}$$

where the complex squeezing parameter $\zeta = re^{i\theta}$ determines the squeezing strength r and the squeezing phase θ [39, 12, 55, 11]. The corresponding single-mode squeezed vacuum state is then given by

$$|\zeta\rangle = \hat{S}(\zeta)\, |0\rangle. \tag{2.7}$$

The Heisenberg picture represents the squeezing transformation that actually elongates and rotates the quadrature axes. In the case of rotated quadrature that are duly rotated Y_1 and Y_2, the variances become

$$V(Y_1) = \tfrac{1}{2} e^{-2r}, \qquad V(Y_2) = \tfrac{1}{2} e^{2r},$$

It is evidently indicating that quantum noise in one quadrature has been reduced below the amount in the vacuum at the cost of increasing the fluctuations in the conjugate quadrature [39].

One of the most important results that were reported in the early papers on quantum-optics is the fact that squeezing implies a *Bogoliubov transformation* (linear canonical transformation) that mixes creation and annihilation operators in the Heisenberg picture [39]:

$$\hat{S}^\dagger(\zeta)\, \hat{a}\, \hat{S}(\zeta) = \hat{a} \cosh r - e^{i\theta} \hat{a}^\dagger \sinh r. \tag{2.8}$$

Such mixing creates *anomalous* (pair) correlations of the form $\langle \hat{a}\hat{a} \rangle$ and $\langle \hat{a}^\dagger \hat{a}^\dagger \rangle$ which vanish for coherent and thermal states. For the squeezed vacuum, the bosonic moments satisfy the well-known relations.

$$\langle \hat{a}^\dagger \hat{a} \rangle = \sinh^2 r, \qquad \langle \hat{a}\hat{a} \rangle = -\frac{1}{2} e^{i\theta} \sinh(2r). \tag{2.9}$$

It is important to point out that the squeezer is coded directly on the level of second-order operator moments [39, 11]. Because Gaussian states are completely characterized by first and second moments, this bosonic moment structure provides an efficient route for treating squeezed states in linear optical networks and open-system models [11, 12].

practically, squeezed states are usually created by using parametric (quadratic) interactions which are naturally bosonic. Typical examples would be degenerate parametric amplification and four-wave mixing, whose Hamiltonian include terms which are proportional to $\hat{a}^{\dagger 2}$ and $\hat{a}^2$ and therefore directly implement the generator appearing in $\hat{S}(\zeta)$ [39]. In integrated photonics and waveguide settings, such bosonic operator descriptions are particularly convenient because propagation, evanescent coupling, and loss can all be modeled as linear transformations and damping channels acting on $\hat{a}_j$ and $\hat{a}_j^\dagger$, allowing one to obtain closed equations for moments such as $\langle \hat{a}_j^\dagger \hat{a}_k \rangle$ and $\langle \hat{a}_j \hat{a}_k \rangle$ [11].

2.1.3.1 Two-Mode Squeezing and Operator Correlations

Beyond single-mode squeezing, the bosonic literature also highlights the importance of *two-mode squeezed states*, generated by the two-mode squeezing operator

$$\hat{S}_{12}(\zeta) = \exp\Big(\zeta^* \hat{a}_1 \hat{a}_2 - \zeta\, \hat{a}_1^\dagger \hat{a}_2^\dagger\Big), \tag{2.10}$$

which creates strong intermodal pair correlations such as $\langle \hat{a}_1 \hat{a}_2 \rangle$ and $\langle \hat{a}_1^\dagger \hat{a}_2^\dagger \rangle$ [11, 12]. These anomalous cross moments are the operator-level signatures of EPR-type correlations and constitute the microscopic origin of Gaussian entanglement and steering when the state is analyzed in the covariance matrix framework [12].

2.1.3.2 Connection to covariance matrices

Whereas the bosonic theory characterizes squeezing through transformations of operators. the transformation interms of bosonic operators is entirely identical to the quadrature covariance matrix description utilized to measure steering and Gaussian entanglement [11, 12]. Realistically, it is common to solve the dynamics in the bosonic basis to get second moments, and then project these moments onto the quadrature covariance matrix which can be easily manipulated and controlled to engage and control in further analysis [11].

2.2 Covariance Matrix

In CV quantum information theory, the covariance matrix provides a compact and practical way to describe the properties of quantum states, especially Gaussian states, which are completely described by their first and second statistical moments [13]. While Gaussian states exist in infinite-dimensional Hilbert spaces, the covariance matrix provides a technically accessible description using a finite number of degrees of freedom [12, 11].

The covariance matrix $\boldsymbol{\sigma}$ is a real, symmetric, and positive-definite matrix of size $2n \times 2n$ for a system of n bosonic modes [12, 11]. It represents the two-mode correlation functions between canonical mode operators [12].

The elements of covariance matrix for n bosonic modes are represented by

$$\sigma_{ij} = \frac{1}{2}\langle \hat{z}_i \hat{z}_j + \hat{z}_j \hat{z}_i \rangle - \langle \hat{z}_i \rangle \langle \hat{z}_j \rangle, \tag{2.11}$$

where $\hat{\mathbf{z}}$ is the phase-space vector and is defined as

$$\hat{\mathbf{z}} = (\hat{Q}_1, \hat{P}_1, \ldots, \hat{Q}_n, \hat{P}_n)^T. \tag{2.12}$$

The degrees of freedom of the bosonic modes are represented by the quadrature operators $\hat{Q}_k$ and $\hat{P}_k$.

Here, $\boldsymbol{\sigma}$ is a real, symmetric matrix that must satisfy the quantum uncertainty principle in order to represent a physical Gaussian state [13].

For a two–mode Gaussian system consisting of modes A and B, the covariance matrix can be written in block form as [12, 11]

$$\boldsymbol{\sigma}_{MN} = \begin{pmatrix} \mathbf{M} & \mathbf{L} \\ \mathbf{L}^T & \mathbf{N} \end{pmatrix}. \tag{2.13}$$

Here, the diagonal blocks, i.e., $\mathbf{M}$ and $\mathbf{N}$ are 2×2 real symmetric matrices describing only local noise and squeezing of each mode, respectively, whereas the off-diagonal blocks $\mathbf{L}$ and $\mathbf{L}^T$ are 2×2 real matrices containing all cross-quadrature moments and therefore fully encode the correlations between the two modes [12]. The covariance matrix in its standard form can be written as

$$\boldsymbol{\sigma}_{MN} = \begin{pmatrix} a & 0 & c_1 & 0 \\ 0 & a & 0 & c_2 \\ c_1 & 0 & b & 0 \\ 0 & c_2 & 0 & b \end{pmatrix}. \tag{2.14}$$

The variances a and b, and the covariance c_1 and c_2 can be determined using the four local symplectic invariants. For example, the determinant of the covariance matrix is $MN = (ab - c_1^2)(ab - c_2^2)$ while the determinants of the local covariance matrices are $\det \mathbf{M} = a^2$ and $\det \mathbf{N} = b^2$, and the determinant of the correlation matrix is $\det \mathbf{L} = c_1 c_2$. Consequently, every arbitrary two–mode covariance matrix admits a unique standard form [4, 11, 12].

A matrix is said to be physically valid covariance matrix, or represents a physical system, if and only if it satisfies the Robertson–Schrödinger uncertainty principle. This condition can be expressed as the matrix inequality [12, 11]

$$\sigma + i\Omega \geq 0, \tag{2.15}$$

where Ω is the symplectic form, defined as the direct sum of n matrices of the form [12, 11]

$$\Omega = \bigoplus_{k=1}^{n} \omega, \qquad \omega = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{2.16}$$

2.3 Optical Waveguides as Quantum Systems

An optical waveguide is a structure that guides light along a desired path. It consists of thin, high-refractive-index core layer positioned between low-refractive-index cladding material and the light is trapped within the core through total internal reflection[67]. Photons passes through these waveguides in particular patterns known as modes [14].

2.3.1 Evanescent Overlap/Coupled Mode Theory

When light passes through a waveguide system, the electric field is mostly trapped in the high-refractive-index core. Yet, a part of the field can penetrate into the surrounding lower-index cladding and this decaying field is known as the evanescent field [66, 67].

If we place another waveguide in this region, i.e., Within the evanescent field, Which typically extends less than 1-2 wavelength, light can tunnel between these two waveguides. We call this transfer of energy through coupling of evanescent field as evanescent coupling. The coupling strenght between the waveguides depends on the separtion distance between the waveguides, refractive index and mode [15].

2.3.2 Loss and Decoherence in Real Waveguide Systems

In real optical waveguide system, the performance is undermined by two processes: classical optical power loss (attenuation) and quantum information loss (decoherence). Exponential decay of optical intensity as light propagates is called attenuation while decoherence is the process in which a quantum system loses its correlation properties i,e superposition or entanglement due to interaction with the environment. This results in losing the necessary phase relationship and limiting visibility of interference [16, 56, 15]. In quantum photonics, the classical optical loss and quantum decoherence are strongly interconnected, that is,the former act as a primary source of the later. When a photon is lost from the waveguide system, its quantum state becomes coupled to unobserved vacuum port which results in transferring quantum information to the environment irreversibly and thus decoheres the entanglement and other nonclassical quantum resources. This loss mechanism can be modelled as an amplitude damping channel [56, 17]

2.3.3 Relevance Of Optical Waveguides For Quantum Circuits/Photonics

The optical waveguides are highly necessary in quantum photonics since they have facilitated the researchers to replace large laboratory systems with small chips that can soon be used in quantum technologies [50, 46]. All structures are built on the chip and this enables the researchers to implement quantum logic gates like CNOT, Toffoli and fredkin gates on a chip. They find practical applications in building quantum computers, secure communication and quantum sensors[61, 46, 50]. Low loss photonic materials like silicon, silicon nitride and lithium niobite are highly essential in this fact since optical waveguides are very important in the character of quantum information processing of integrated photonic circuits. Their nonlinear and optical properties can be optimally used in control of generation, manipulation, and detection of quantum states with very low propagation loss. When photon sources, beam spliters, phase shifters, and detectors are implemented on a single chip on a waveguide basis the resultant circuit is compact, stable, and scalable [17, 18].

2.4 Quantum Steering

The phenomenon known as quantum steering was first introduced in 1935 [8]. It is a form of quantum correlation that has received considerable attention in quantum information theory [38]. Schrodinger first described this phenomenon as "magic" as the effect does not facilitate transmission of information, and it is the ability of one party (Alice) to measure or steer the wave function of another distant party [19]. One common notion in the modern theory is the inability of the conditional quantum states of one party (Bob) to be described by a local hidden state (LHS) model. A joint bipartite quantum state ϱ_{AB} is said to be steerable if Bob is forced to believe that Alice is influencing his system through some form of action at a distance. Conversely, when (LHS) exists for the shared state, then such a bipartite system is referred to as an unsteerable state [19, 20]. Steering lies between entanglement and Bell nonlocality, that is, only steerable quantum correlations can be verified in one-sided device independent scenario, entanglement cannot [19, 20]. In CV systems, generally implemented using optical field quadratures Q and R, the steering-detection problem is often formulated in terms of criteria based on the Heisenberg uncertainty relation for conditional measurements. A state shared between A and B is steerable if the product of the minimum inferred variances of Bob's position position (Q_B) and momentum (R_B) quadratures, in terms of the measurements of Alice, is lower than the quantum limit: [21, 19]

$$V_{\text{inf}}(Q_B)\, V_{\text{inf}}(P_B) < 1. \tag{2.17}$$

For CV Gaussian states where both the states and measurements are Gaussian, this EPR criterion is both necessary and sufficient to detect bipartite steering[21]. Remarkably, a bipartite Gaussian state is unsteerable with Gaussian measurements if and only if a specific matrix inequality involving the covariance matrix V_{AB} is satisfied:

$$V_{AB} + i\,(0_A \oplus \Omega_B) \geq 0. \tag{2.18}$$

If a quantum state is steerable with Gaussian measurements, it is also steerable with a certain pair of canonical quadratures [19].

2.4.1 Quantification of Quantum Steering using Gaussian Steering Measure

The Gaussian $1 \to 2$ steerability $\mathcal{S}_{1\to 2}(\sigma_{MN})$ is defined based on the symplectic eigenvalues $\{\bar{\nu}_j^N\}$ of the Schur complement matrix M_{σ_N}, calculated from the state's covariance matrix σ_{MN}:

$$\mathcal{S}_{1\to 2}(\sigma_{MN}) := \max\left\{0,\ -\sum_{\bar{\nu}_j^N<1} \ln\left(\bar{\nu}_j^N\right)\right\}. \tag{2.19}$$

For the specific case of two-mode Gaussian states (where the steered party has only one mode), the measure simplifies to the Rényi-2 coherent information:

$$\mathcal{S}_{1\to 2}(\sigma_{MN}) = \max\left\{0,\ \frac{1}{2}\ln\left(\frac{\det\sigma_M}{\det\sigma_{MN}}\right)\right\} = \max\left\{0,\ H_2(\sigma_M) - H_2(\sigma_{MN})\right\}, \tag{2.20}$$

where $H_2(\sigma)$ denotes the Rényi-2 entropy [22].

2.5 Quantum Entanglement In Continuous Variable Systems

Entanglement is a fundamental property of a quantum mechanics, and it is a central resource for most quantum information processing protocols and emerging quantum technologies [23]. Originating from conceptual structures identified by Einstein, Podolsky, and Rosen (EPR) and Schrödinger, entanglement describes nonlocal correlations within composite quantum systems that cannot be replicated by classical systems [25]. It is a physical fact that two or more particles show correlations which cannot be explained classically, and these particles will be correlated no matter how far they are from each other [1, 25]. Historically considered a "spooky" phenomenon, entanglement is now an important part of engineering essential quantum protocols, such as, quantum computing and quantum teleportation in CV systems associated with infinite dimensional Hilbert spaces, provide the advantage of allowing deterministic and unconditional quantum operations of which is often a shortcoming in DV schemes due to relying on an imperfect single-photon generation and detection [23]. Within the CV framework, the family of Gaussian states is mostly used for the generation and manipulation of entangled states [26]. The quantitative way to determine the intensity or the extent of entanglement within a quantum system is termed as entanglement quantification. Quantification is needed and important because the measured degree of entanglement directly influences the performance of quantum information tasks [1, 25, 33].

2.5.1 Entanglement Quantification Measures

To measure the entanglement created in a system in a quantitative way, scientists adopted various measures[1] . One of the most commonly used measure is logarithmic negativity E_N.

E_N [57, 58], which is defined as

$$E_N(t) = \log_2 \left\| \rho_{MN}^{T_N} \right\|_1 , \tag{2.21}$$

where ρ_{MN} denotes the density matrix of the bipartite system M and N, $\rho_{MN}^{T_N}$ denotes the partial transpose with respect to subsystem N, and $\|\cdot\|_1$ represents the trace norm. A non-zero value for E_N confirms that the state is entangled.

For a two mode Gaussian state, the measure of entanglement, E_N, is determined by the symplectic eigenvalues ($\tilde{\nu}$) of the partially transposed covariance matrix ($\tilde{\sigma}_{MN}$). The state is entangled if the smallest symplectic eigenvalue, $\tilde{\nu}$ satisfies

$$\tilde{\nu} < \frac{1}{2}. \tag{2.22}$$

Logarithmic negativity in this context is given by

$$E_N(t) = \max\left[0, \; -\ln\left(2\,\tilde{\nu}(t)\right)\right]. \tag{2.23}$$

and serves as an upper bound to the distillable entanglement [57, 58, 28] . Here, $\tilde{\nu}$ is the minimum eigen value of the covariance matrix.

Properties and Hierarchy: The Gaussian steering measure satisfies several desirable mathematical properties, for example, it is invariant under local unitaries, convex, additive, and monotonically non-increasing under quantum operations performed on the untrusted party (Alice). It vanishes if and only if the state is nonsteerable by Gaussian measurements. Moreover, quantum entanglement and steering satisfy strict hierarchy relation, such that

$$\mathcal{S}_{1\to 2}(\sigma_{MN}) \leq \mathcal{E}_2(\sigma_{MN}), \tag{2.24}$$

where $\mathcal{E}_2$ denotes the Gaussian Rényi-2 measure of entanglement. For pure states,

$$\mathcal{S}_{1\to 2}(\sigma^p_{MN}) = \mathcal{E}_2(\sigma^p_{MN}). \tag{2.25}$$

2.6 Loss and Decoherence

Decoherence has become a prominent subject in modern physics, It is motivated by the desire to understand the border between the classical and quantum world, and the manner in which it is detrimental to the processing of quantum information [29]. Fundamentally, decoherence is nothing but the irreversible and uncontrollable accumulation of quantum correlations, such as entanglement and steering, between a system and its environment [29, 30]. Decoherence has attracted a lot of interest because of its relevance in explaining the quantum-classical boundary as well as its harmful effects on quantum information processing technologies. The main mechanism is that the environment actively monitors some system observable and hence, the decoherence of non-preferred states takes place. This monitoring thus imposes the choice of pointer states, which are robust against environmental interactions[31, 29]. These pointer states continue to correlate with their environment while remaining stable under environmental interactions. The rapid conversion of quantum superposition to mixtures is the reason why superposition states, which are commonly known as "Schrodinger cat" states, do not seem to appear. The selection of such preferred states is known as Environment-Induced Superselection (Einselection). Einselection places a basic limitation on the bulk of the system's Hilbert space, causing it to be restricted to a small classical domain that is made up of pointer states [31].

2.6.1 Loss (Dissipation) and Pure Decoherence

Decoherence is usually accompanied by dissipation, which refers to the net exchange of energy with the environment [29]. But there are certain theoretical cases that focus on pure decoherence (or dephasing), in which there is no energy exchange with the environment. The phenomenon of pure decoherence is of special interest in explaining why macroscopic superposition do not occur, since the mechanism that leads to this is proposed to rely on the distinguishability of quantum states rather than on the differences in their energy [29]. Pure decoherence can occur when the system Hamiltonian (H_S) approximately commutes with the system-bath interaction Hamiltonian (H_{int}), or when the system initial states vary at a rate significantly slower than the decoherence timescale. Pure decoherence is considered very important in theoretical models which attempt to explain the apparent lack of superposition of macroscopically distinguishable states, including Schrodinger-cat states [29]. The roles of these phenomena should be differentiated: relaxation and noise (loss) are the processes in which the environment perturbs the system, decoherence and einselection are the processes in which the system perturbs the environment [31]. Another extremely significant thing about decoherence, particularly in macroscopic systems, is the separation of the time scales between decoherence and energy loss (relaxation). The rate at which a system decoheres is often depends on the separation between the superposed, macroscopically distinct states. In such cases, the dimensionless distance D can be very large (e.g., 10^{20}), so decoherence can be very fast, even when the rate of energy relaxation of the system is very small [32].

2.6.2 Loss Mechanism in Waveguides

Loss mechanism (κ) in waveguides is due to the loss in the material of the waveguide structure itself. Waveguides have often been designed, to be identical, and so the loss rate for the field modes ($a1$ and $a2$) is usually taken to be the same. Loss is theoretically modelled in the framework of system–reservoir interaction by using a quantum-Liouville equation of motion. This is the equation of evolution of the density operator (ρ) with two terms, the Hamiltonian term to the unitary coupling and the loss term ($L\rho$) to represent the dissipation:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + L\rho, \tag{2.26}$$

where the loss operator $L\rho$ specifically represents the decay rates of the modes $a1$ and $a2$ [28, 56, 54, 55] .

2.7 Applications and Motivations

In this section, we discuss the applications and motivations of quantum entanglement, quantum steering, and optical waveguides.

2.7.1 Applications of Quantum Entanglement

The concept of quantum entanglement is a central pillar of quantum information theory and quantum mechanics and is used to explain highly correlated particles that become linked regardless of the distance separating them [1, 25]. This has a critical role in proving the theoretical and algorithmic superiority of quantum computers over its classical counterparts [34]. An important element of quantum computing is the manipulation of entanglement that results in the enhancement of the capabilities [34], and is applied in fundamental quantum information tasks like quantum cryptography, quantum teleportation and quantum computing [1]. The quantification of entanglement is important in the determination of the intactness and the depth of the level of correlation that has a direct impact on the execution of quantum information tasks. Quantified entanglement measures include concurrence, negativity (Including logarithmic negativity), Schmidt decomposition, and entanglement of formation [25]. In case of bipartite Gaussian states, entanglement can be quantitatively measured by the logarithmic negativity. In waveguide architecture, the sustainability of generated entanglement can be significantly important in practical implementation of quantum communication and computation tasks [28]. Highly entangled states are essential in doing complex quantum information processing and implementing error-correcting code [34]. In short quantum entanglement is essential for achieving quantum computational advantage and quantum error-correcting codes, it allows safe quantum communication protocols, scalable on-chip implementations in integrated optical waveguide architectures [1, 25, 34, 11, 23, 28].

2.7.2 Applications of Quantum Steering

Quantum steering is a form of quantum correlation that has been introduced by Schrodinger which allows one party, Alice, to change the state of a remote party, Bob, by performing some local measurements which make use of shared entanglement [8, 38, 19]. One of the most practical uses of quantum steering is the realization of desired quantum states, the so-called measurement-Induced quantum Steering (MIQS), a protocol that can be used to prepare a target quantum state regardless of the initial (mixed) state [34]. Quantum Steering plays an important role in offering security to one-sided device-independent quantum key distribution (1SDI QKD) protocols, where the amount of steerability (Gaussian) can be used to measure the rate of guaranteed key rate [22].

Quantum steering also has a close connection with quantum teleportation, any Gaussian state that can perform safe teleportation must be two way steerable [35]. Quantum steering also plays an important role in quantum-enhanced metrology [36]. Hence, quantum steering has important applications in measurement-induced quantum state preparation, one-sided device-independent quantum key distribution, secure CV quantum teleportation, quantum cryptography, and quantum-enhanced metrology. In these protocols, steerable can be used to accomplish tasks that cannot be certified using entanglement alone [38, 19, 22, 35, 36].

2.7.3 Applications of Waveguides

Coupled waveguide designs are significant fundamental components that scientists examine whether they can be employed as straightforward building blocks for quantum circuits in integrated photonic devices[47]. These are discrete optical systems that are quite efficient in controlling the movement of light. Waveguides allowed a CNOT gate on a single silicon chip, which are useful in executing key activities in quantum computing. Such architectures form a basis for universal photonic quantum circuits and are relevant for the development of photonic implementations of quantum algorithms, including circuits related to Shor's quantum factoring algorithm [46, 61, 50]. Beyond computing, waveguide arrays are used to investigate coherent phenomena such as quantum walks and a discrete form of the Talbot effect [28, 59, 46, 60]. In short, with integrated optical Waveguides, it is now possible to implement quantum logic gates and universal photonic quantum circuits, they give rise to on-chip implementations of quantum algorithms, including Shor-factoring, and enable quantum simulation by coherent phenomena such as quantum walks and Talbot effects [61, 50, 46, 47, 62], they also provide a scalable environment to study the propagation and stability of nonclassical states in realistic lossy and open quantum systems [59, 46, 60, 28].

In this chapter, we discuss physical model of our system and construct the theoretical framework which is required to study its quantum dynamics.

Chapter 3

Model System and Theoretical Formulations

3.1 The Model System

Our proposed system consists of two identical single-mode waveguides which are placed parallel to each other in a close distance such that their evanescent field overlap and enables coupling between them. The optical modes in each waveguide are represented by creation and annihilation operators a_1 and a_2, where a_1 represents optical mode in the first waveguide and a_2 represents optical mode in the second waveguide. Each waveguide mode act as a bosonic harmonic oscillator. It is to be noted that the two waveguides are assumed to be identical and at the same resonance frequency and with the same loss rate, which is a realistic assumption.

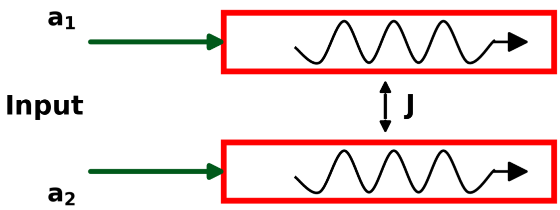

Figure 3.1: *Schematic diagram of a coupled waveguide system.*

3.2 Hamiltonian of the Coupled Waveguide System

In this section, we discuss our system Hamiltonian, and coupling between waveguide modes.

3.2.1 System Hamiltonian

The Hamiltonian for our system is given by

$$H = \hbar\omega \left(a_1^\dagger a_1 + a_2^\dagger a_2 \right) + \hbar J \left(a_1^\dagger a_2 + a_2^\dagger a_1 \right). \tag{3.1}$$

Here, a_1 and a_2 are the bosonic field operators for the optical modes in each waveguide, satisfying the commutation relations $[a_1, a_1^\dagger] = 1$ and $[a_2, a_2^\dagger] = 1$.

The symbol ω is angular frequency of each waveguide mode and the parameter J in the second part of the Hamiltonian represents the coupling strength between the two waveguides and its value depends on the separation distance between the waveguides.

The Hamiltonian consists of two main parts, that is, the free energy part and the coupling part. The first part, $\hbar\omega\left(a_1^\dagger a_1 + a_2^\dagger a_2\right)$, correspond to the free energy of the individual waveguide modes. The second part, account for the evanescent coupling between the two waveguide modes, $\hbar J\left(a_1^\dagger a_2 + a_2^\dagger a_1\right)$, accounts for the evanescent coupling between the two waveguide modes.

3.2.2 Physical Interpretation of the Coupling

The evanescent interaction between the two parallel optical waveguides is represented by the coupling term in the system Hamiltonian. When the waveguides are sufficiently close to each other, the electromagnetic field of the guided mode extends beyond the core of the waveguide and overlaps with the field of the neighboring waveguide. This overlap allows the coherent exchange of photons between the two modes and is called as evanescent overlapping [15, 63].

The strength of the coupling between the waveguides is defined by the parameter J which defines the speed with which energy moves between the waveguides. The higher the value of J the higher the mode overlap and higher the speed of oscillatory exchange of photons. When the two waveguides are far away from each other such that the coupling parameter is zero, they are entirely independent and are considered uncoupled. Physically, such an interaction is similar to a beam splitter, where the photons coherently and simultaneously hopped between the two modes without loss in ideal case [15, 28, 46].

Without the dissipation, the coherent exchange between the two waveguides results in a periodic oscillation of photon between the two waveguides. In case of losses, such oscillations are slowly damped, although the coupling term still plays the role of creating and maintaining quantum correlations. The coupling is therefore the core of creation maintenance of quantum interactions in integrated photonic systems [28, 15].

3.3 Open-System Dynamics and Loss Modelling

In realistic integrated photonic systems, optical modes experience losses. The coupled waveguide system cannot be treated as an isolated system, it needs to be formulated in the context of an open quantum systems. To include these dissipative effects, the system dynamics is determined using the density matrix formalism, in which the interaction with the external environment results in decoherence and energy decay [56, 28]

.The time evolution of the coupled waveguide system along with the losses factors is formulated interms of the Lindblad form of the quantum Liouville (master) equation. This approach allows inclusion of photon loss without changing the nature of the Gaussian states. This preservation is necessary for the analysis of quantum steering and entanglement [56, 54, 11, 28].

3.3.1 Quantum Liouville (Master) Equation

The time evolution of the proposed system with loss terms is determined using the following quantum liouville equation [56, 54].

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \mathcal{L}\rho. \tag{3.2}$$

In the above equation, the first part of the commutator generates the time evolution, where ρ represents density matrix of the system, which generates time evolution and is responsible for oscillatory exchange of energy and quantum correlation between the waveguides due to J. However, the Lindblad term ($L\rho$) deals with the losses, decoherences and decays due to interaction of system with the environment [56, 28].

$$\begin{aligned}\mathcal{L}\rho = &-k_1\left(a_1^\dagger a_1\rho + \rho a_1^\dagger a_1 - 2a_1\rho a_1^\dagger\right)\\ &- k_2\left(a_2^\dagger a_2\rho + \rho a_2^\dagger a_2 - 2a_2\rho a_2^\dagger\right),\end{aligned} \tag{3.3}$$

Here, k_1 and k_2 represent the loss rates of the first and second modes, and a_i and $a_i^\dagger$ (with $i = 1, 2$) are the annihilation and creation operators associated with the two modes [56, 28].

3.4 Dynamical Equations of the Model System

This section deals with the derivation of the dynamical equations of the coupled waveguide system by evaluating the expectation values, the open-system dynamics can be expressed as in terms of second-order moments, which is sufficient to describe the Gaussian states at the output of the waveguides. In this context, the time evolution of an arbitrary observable O is obtained using the following general expression [56].

$$\frac{d}{dt}\langle O\rangle = \mathrm{Tr}(O\dot{\rho}) = -i\langle[O,H]\rangle + \langle\mathcal{L}^\dagger(O)\rangle. \tag{3.4}$$

Hence, using Eq. (3.4) along with Eqs. (3.2) and (3.3),

the equations of motion for the relevant second-order moments are obtained as

$$\frac{d}{dt}\langle a_1^\dagger a_1\rangle = iJ\Big(\langle a_2^\dagger a_1\rangle - \langle a_1^\dagger a_2\rangle\Big) - k_1\langle a_1^\dagger a_1\rangle, \tag{3.5}$$

$$\frac{d}{dt}\langle a_2^\dagger a_2\rangle = iJ\Big(\langle a_1^\dagger a_2\rangle - \langle a_2^\dagger a_1\rangle\Big) - k_2\langle a_2^\dagger a_2\rangle, \tag{3.6}$$

$$\frac{d}{dt}\langle a_1^\dagger a_2\rangle = iJ\Big(\langle a_2^\dagger a_2\rangle - \langle a_1^\dagger a_1\rangle\Big) - \frac{k_1+k_2}{2}\langle a_1^\dagger a_2\rangle, \tag{3.7}$$

$$\frac{d}{dt}\langle a_2^\dagger a_1\rangle = iJ\Big(\langle a_1^\dagger a_1\rangle - \langle a_2^\dagger a_2\rangle\Big) - \frac{k_1+k_2}{2}\langle a_2^\dagger a_1\rangle, \tag{3.8}$$

$$\frac{d}{dt}\langle a_1 a_1\rangle = -2iJ\langle a_1 a_2\rangle - k_1\langle a_1 a_1\rangle, \tag{3.9}$$

$$\frac{d}{dt}\langle a_2 a_2\rangle = -2iJ\langle a_1 a_2\rangle - k_2\langle a_2 a_2\rangle, \tag{3.10}$$

$$\frac{d}{dt}\langle a_1 a_2\rangle = -iJ\Big(\langle a_1 a_1\rangle + \langle a_2 a_2\rangle\Big) - \frac{k_1+k_2}{2}\langle a_1 a_2\rangle. \tag{3.11}$$

These equations form a closed set due to the quadratic structure of the Hamiltonian and the Gaussian-preserving nature of the Lindblad dissipation. Although seven coupled moment equations are obtained, and the mode Coherence moments, appearing in Eqs. (3.7) and (3.8), satisfy the Hermitian relation

$$\langle a_2^\dagger a_1\rangle = \langle a_1^\dagger a_2\rangle^*, \tag{3.12}$$

the system dynamics can be fully described by six coupled second-order moments.

3.5 Analytical Solutions of the Moment Equations

In this section, we present the closed-form solutions of the moment differential equations derived in the previous section. These solutions represents the time evolution of photon number and coherence correlations between the coupled waveguide modes in the presence of losses. In order to solve the dynamical rate equations, given in Eqs. (3.5) to (3.11), we use the following initial conditions:

$$n_1 = \langle a_1^\dagger a_1 \rangle + \frac{1}{2}, \quad n_2 = \langle a_2^\dagger a_2 \rangle + \frac{1}{2}, \quad m_1 = \langle a_1 a_1 \rangle, \quad m_2 = \langle a_2 a_2 \rangle,$$

and

$$c(0) = s(0) = 0.$$

For identical waveguides, we assume equal loss rates,

$$k_1 = k_2 = \kappa. \tag{3.13}$$

For our convenience, we use the following notations for the moments:

$$n_1(t) = \langle a_1^\dagger a_1 \rangle(t), \quad n_2(t) = \langle a_2^\dagger a_2 \rangle(t),$$

$$c(t) = \langle a_1^\dagger a_2 \rangle(t), \quad m_1(t) = \langle a_1 a_1 \rangle(t), \quad m_2(t) = \langle a_2 a_2 \rangle(t), \quad s(t) = \langle a_1 a_2 \rangle(t).$$

Note that, the term $\frac{1}{2}$ accounts for the unavoidable vacuum (zero-point) fluctuations of the bosonic field. For independently prepared input modes, the initial inter-mode correlations vanish,

$$\langle a_1^\dagger a_2 \rangle_0 = \langle a_1 a_2 \rangle_0 = 0,$$

leading to simplified expressions that are used throughout the remainder of this thesis. Hence, the exact solutions of the moment equations are given by,

$$n_1(t) = \frac{e^{-\kappa t}}{2} \Big[(n_1 + n_2 - 1) + (n_1 - n_2) \cos(2Jt) \Big], \tag{3.14}$$

$$n_2(t) = \frac{e^{-\kappa t}}{2} \Big[(n_1 + n_2 - 1) - (n_1 - n_2) \cos(2Jt) \Big], \tag{3.15}$$

$$c(t) = \frac{i\, e^{-\kappa t}}{2} (n_2 - n_1) \sin(2Jt), \tag{3.16}$$

$$m_1(t) = \frac{e^{-\kappa t}}{2} \Big[(m_1 + m_2) \cos(2Jt) + (m_1 - m_2) \Big], \tag{3.17}$$

$$m_2(t) = \frac{e^{-\kappa t}}{2} \Big[(m_1 + m_2) \cos(2Jt) + (m_2 - m_1) \Big], \tag{3.18}$$

$$s(t) = -\frac{i\, e^{-\kappa t}}{2} (m_1 + m_2) \sin(2Jt). \tag{3.19}$$

These expressions can be directly used in the covariance matrix for the numerical evaluation of the Gaussian quantum steering and entanglement.

3.6 Covariance Matrix and Quantification of Quantum Steering and Entanglement

A SMGS is fully characterized by the parameters of its covariance matrix. The diagonal element n_j quantifies the local quadrature variance of the j-th mode and is directly related to the mean photon number according to

$$\langle a_j^\dagger a_j \rangle_0 = n_j - \tfrac{1}{2}, \tag{3.20}$$

where the term $\frac{1}{2}$ accounts for the unavoidable vacuum (zero-point) fluctuations of the bosonic field [11].

Within the nonclassicality–purity parametrization of SMGS, the same quantity n_j can be expressed in terms of the nonclassicality τ_j and the purity μ_j as [37]

$$n_j = \frac{1}{2} + \frac{\tau_j^2 + \dfrac{1}{(2\mu_j)^2} - \dfrac{1}{4}}{1 - 2\tau_j}. \tag{3.21}$$

which ensures that the corresponding covariance matrix satisfies the Heisenberg uncertainty principle [11].

The off-diagonal covariance parameter m_j is defined through the anomalous second-order moment

$$\langle a_j a_j \rangle_0 = -m_j, \tag{3.22}$$

where $m_j = |m_j| e^{i\phi_j}$, in general, is a complex quantity within the same nonclassicality–purity parametrization of SMGS, and

$$|m_j| = \frac{\tau_j - \tau_j^2 + \dfrac{1}{(2\mu_j)^2} - \dfrac{1}{4}}{1 - 2\tau_j}. \tag{3.23}$$

These two parameters, n_j and m_j, together fully specify the single-mode Gaussian input state, with n_j determining the phase-insensitive fluctuations and m_j encoding the phase-sensitive correlations of the field [37].

3.6.0.1 Covariance Matrix of the Two-Mode System

In general, the covariance matrix for a two-mode system is given by

$$\boldsymbol{\sigma}(t) = \begin{pmatrix} \mathbf{M}(t) & \mathbf{L}(t) \\ \mathbf{L}^{\dagger}(t) & \mathbf{N}(t) \end{pmatrix}. \tag{3.24}$$

In the above matrix, the 2×2 submatrices M and N, represents the local variations in the first and second modes, while the 2×2 submatrix L represents the correlation between the modes. Explicitly, these block matrices are given by

$$\mathbf{M}(t) = \begin{pmatrix} n_1(t) + \frac{1}{2} & -m_1(t) \\ -m_1^*(t) & n_1(t) + \frac{1}{2} \end{pmatrix}. \tag{3.25}$$

$$\mathbf{N}(t) = \begin{pmatrix} n_2(t) + \frac{1}{2} & -m_2(t) \\ -m_2^*(t) & n_2(t) + \frac{1}{2} \end{pmatrix}. \tag{3.26}$$

$$\mathbf{L}(t) = \begin{pmatrix} c^*(t) & -s(t) \\ -s^*(t) & c(t) \end{pmatrix}. \tag{3.27}$$

The covariance matrix in Eq. (3.24) fully characterizes the Gaussian state, and provides the basis for identifying different quantum correlations [10, 11, 12]. This block-structured covariance matrix serves as the starting point for evaluating both quantum entanglement and quantum steering.

3.7 Gaussian Quantum Steering

Gaussian quantum steering is determined through the conditional covariance matrix of one mode, which is obtained by performing measurements on the other mode. Mathematically, this can be done by taking the Schur complement of the total covariance matrix.

The degree of Gaussian steering from mode 1 to mode 2, with a covariance matrix σ, is quantified by [22, 40, 26, 28].

$$S_{1\to 2} = \max\left[0,\ \frac{1}{2}\log_2 \frac{\det \mathbf{N}}{\det \boldsymbol{\sigma}_{2|1}}\right]. \tag{3.28}$$

3.8 Gaussian Quantum Entanglement

With the covariance matrix at hand, bipartite quantum entanglement can be easily detected by performing the partial transpose operation. However, to quantify the degree of entanglement, logarithmic negativity is one of the best quantifiers, which is given by

$$E_N = \max[0, -\log_2(2\tilde{\nu}_-)]\,, \tag{3.29}$$

where, ν_- is the symplectic eigenvalue of the covariance matrix. This quantifies the degree of entanglement between the two subsystems (modes) [28].

For a TMGS, the smallest symplectic eigenvalue $\tilde{\nu}_-$ of the partially transposed covariance matrix is given by [13, 10, 28]

$$\tilde{\nu}_- = \sqrt{\frac{\tilde{\Delta} - \sqrt{\tilde{\Delta}^2 - 4\,\det \boldsymbol{\sigma}(t)}}{2}},$$

where

$$\tilde{\Delta} = \det \mathbf{M}(t) + \det \mathbf{N}(t) - 2\det \mathbf{L}(t).$$

3.9 Relation Between Entanglement and Steering

Although quantum and quantum steering can both be determined from the same covariance matrix, they explore different aspects of quantum correlations. Entanglement is detected by the partial transpose of the covariance matrix and is symmetric between the subsystems, whereas steering is detected through conditional uncertainty relation and is intrinsically directional. All the steerable Gaussian states are therefore entangled, but not all entangled states exhibit steering [19, 38] .

The quantum hierarchy relation between quantum correlations is essential and the same strict hierarchy holds for continuous-variable Gaussian states [38]:

$$\text{Bell nonlocality} \;\subsetneq\; \text{Gaussian Steering} \;\subsetneq\; \text{Gaussian Entanglement}. \tag{3.30}$$

Since, the focus of this thesis is on generation of quantum steering and entanglement, therefore, we are going to show the hierarchy relation, such that,

$$\text{Gaussian Steering} \subsetneq \text{Gaussian Entanglement}. \tag{3.31}$$

This relation simply means that quantum steering is a subset of quantum entanglement [19].

Chapter 4

Results and Discussion

In this chapter, we study the dynamical behaviour of the Gaussian quantum steering and entanglement in a coupled lossy waveguide system and analyze the graphical results. We initially analyze by considering a general reference case where the nonclassicality, purity and loss parameters are held constant and then by considering the action of each of these parameters separately. We investigate the effect of these parameters by plotting quantum entanglement and quantum steering against dimensionless interaction time $\frac{Jt}{\pi}$. Furthermore, we consider a special kind of Gaussian states called squeezed states and analyzed the effect of squeezing on quantum entanglement and quantum steering. Finally, we investigate effect of coupling parameter J on quantum entanglement and quantum steering by plotting quantum entanglement and quantum steering against κt.

4.1 Time Evolution of Gaussian Quantum Steering and Entanglement

This section presents the analysis of the general time dynamics of the Gaussian quantum steering and entanglement in a coupled lossy waveguide system. The nonclassicality τ, purity μ, and loss parameter κ are kept fixed, and the time evolution of entanglement and steering is investigated. The evolution of the logarithmic negativity $E_{\mathcal{N}}$ and the Gaussian quantum steering $\mathcal{S}$ with the normalized propagation time is illustrated by Fig 4.1. From the figure, we see that the quantum correlations are periodic, which means there is coherent exchange between quantum correlations of the two interacting waveguide modes. The maxima and minima of Gaussian steering and entanglement are situated in the same propagation lengths meaning that both the correlations can be related to the same underlying mode coupling dynamics that is brought about by the evanescent interaction between the waveguides. It is also noted that the value of Gaussian quantum steering is never greater than that of the Gaussian quantum entanglement, which means the modes in waveguides satisfy quantum hierarchy.

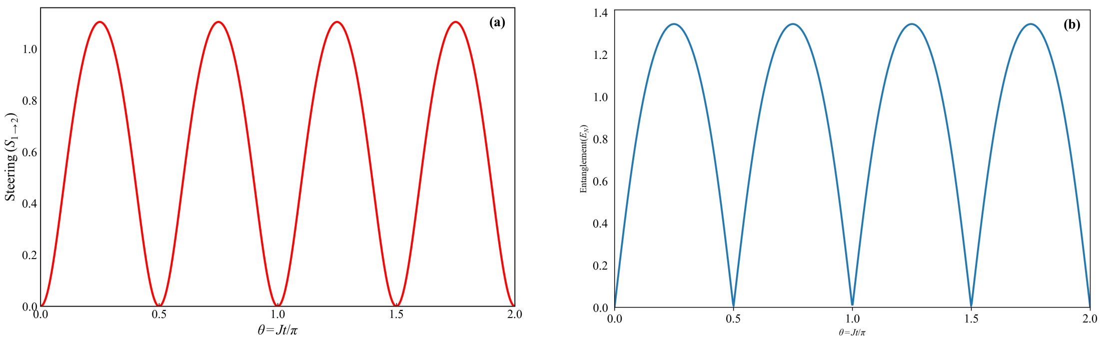

Figure 4.1: *Two kinds of quantum correlations (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time, for fixed parameters* $\mu_1 = \mu_2 = \mu = 1$, $\tau_1 = \tau_2 = \tau = 0.35$, $\kappa_1 = \kappa_2 = \kappa = 0$ *and* $\phi_1 = 0$ *and* $\phi_2 = \pi/2$.

4.2 Effect of Nonclassicality τ

In this section, we analyze the effect of nonclassicality τ on Gaussian quantum steering and entanglement, where we choose other parameters, i.e., the purity μ and the loss parameter κ fixed in this analysis. Fig 4.2 indicates that for the higher values of the nonclassicality both, quantum entanglement and quantum steering becomes stronger i.e., the larger the τ the larger the peaks, so quantum steering and entanglement are larger.

This is to imply that when the nonclassicality of the initial Gaussian state is greater, their ability to generate and maintain quantum correlations in the coupled waveguide system is also enhanced. In addition, the Gaussian steering responds more to variation in the nonclassicality in comparison to entanglement. The magnitude of quantum entanglement is greater than quantum steering for all values of τ, hence strongly following the quantum hierarchy relation.

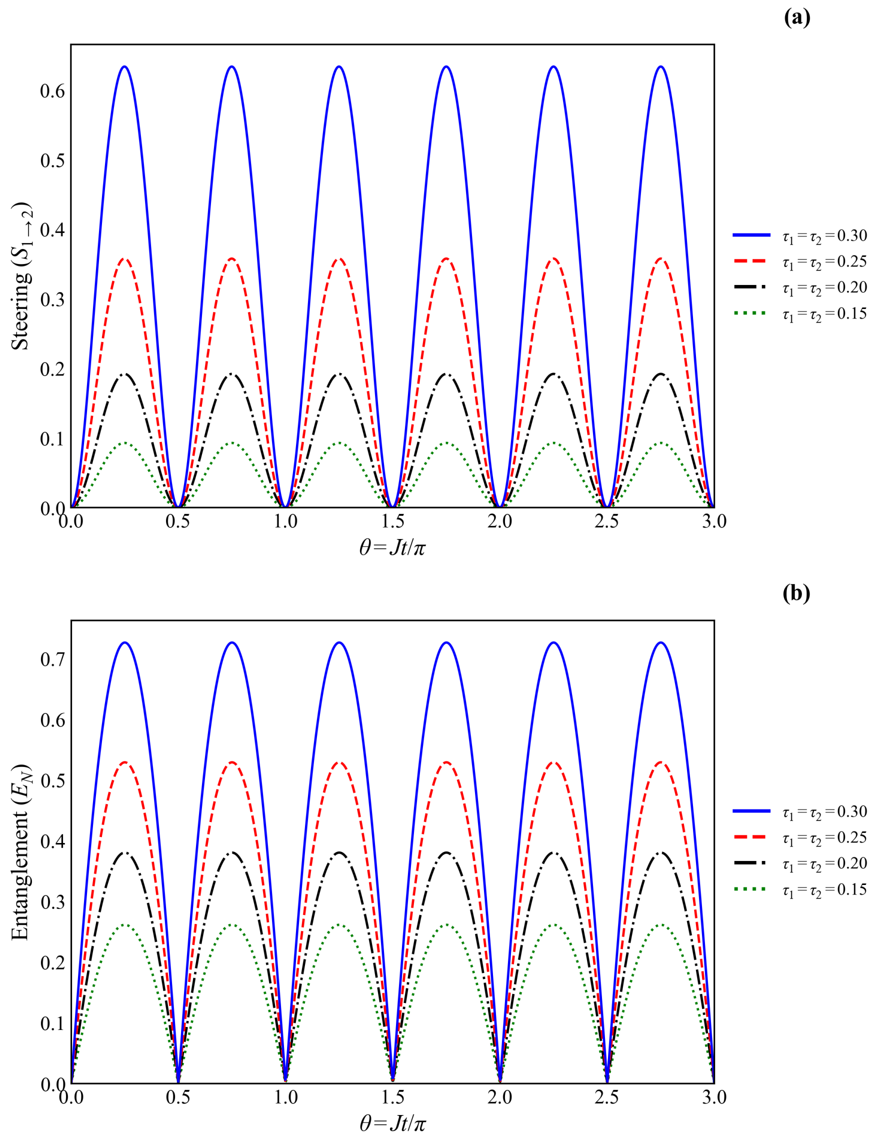

Figure 4.2: *Two kinds of quantum correlations: (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time, for fixed parameters* $\kappa_1 = \kappa_2 = \kappa = 0.05$, $\mu_1 = \mu_2 = \mu = 1$, $\phi_1 = 0$ *and* $\phi_2 = \pi/2$. *In each case, dotted-green, dot-dashed black, dashed-red, and solid blue lines represent* $\tau_1 = \tau_2 = \tau = 0.15$, 0.20, 0.25, *and* 0.30, *respectively.*

4.3 Effect of Purity μ

In this section, we examine the impact of purity μ of the input state on the development of Gaussian quantum steering and entanglement as plotted in Fig. 4.3. This figure demonstrates that when we reduce the value of the purity μ, we observe that, both quantum correlations i.e., quantum steering and quantum entanglement decreases, indicating that these are vulnerable to noise and decoherence.

Moreover, the magnitude of quantum entanglement is greater than quantum steering for all values of μ, hence strictly following the hierarchy relation.

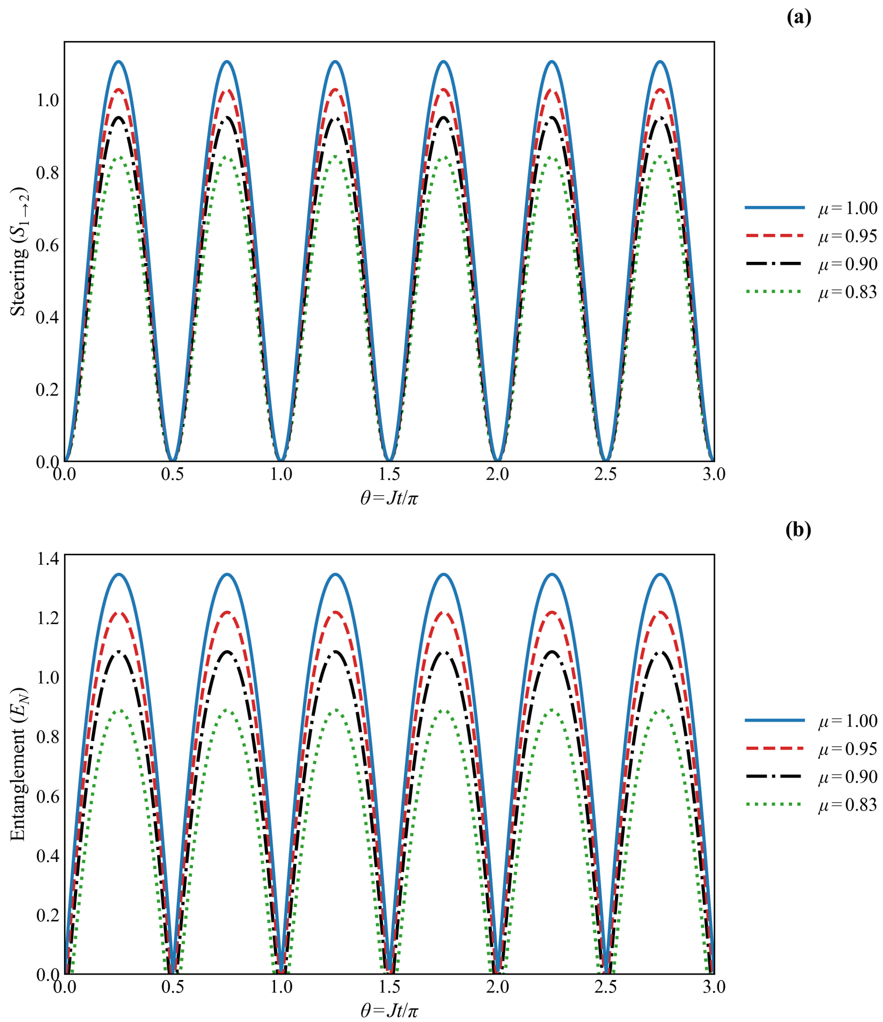

Figure 4.3: *Two kinds of quantum correlations: (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time, for fixed parameters $\kappa_1 = \kappa_2 = \kappa = 0.05$, $\tau_1 = \tau_2 = \tau = 0.35$, $\phi_1 = 0$ and $\phi_2 = \pi/2$. In each case, dotted-green, dot-dashed black, dashed-red, and solid blue lines represent $\mu_1 = \mu_2 = \mu = 0.80$, 0.90, 0.95, and 1.0, respectively.*

4.4 Effect of Loss Parameter κ

In this section, we investigate how the generation of Gaussian quantum steering and entanglement via a coupled waveguide affect when the loss parameter κ is changed. In this connection, the two types of quantum correlations are plotted against dimensionless interaction time, as shown in Fig. 4.4. The result in the figure clearly indicates that quantum correlations decrease significantly as the loss parameter increases. Moreover, we can see that both the strength and time development of quantum steering is less than quantum entanglement for all values of κ. Hence, quantum steering and entanglement satisfy the hierarchy relation both in magnitude and time development

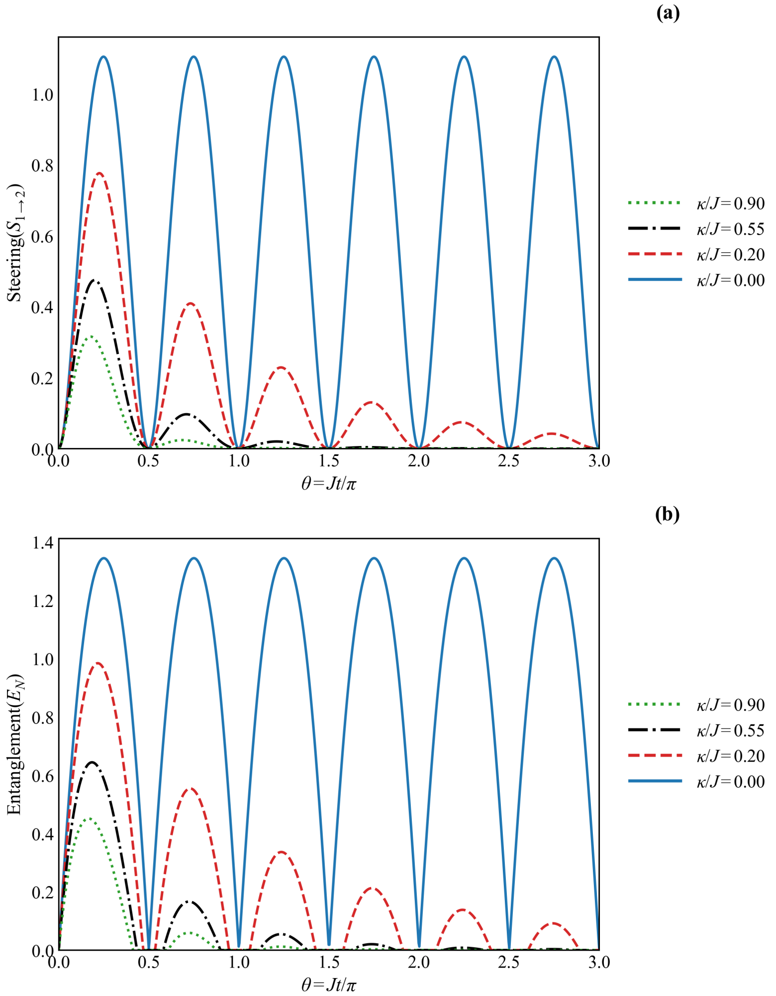

Figure 4.4: *Two kinds of quantum correlations: (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time, for fixed parameters $\tau_1 = \tau_2 = \tau = 0.35$, $\phi_1 = 0$ and $\phi_2 = \pi/2$, and $\mu_1 = \mu_2 = \mu = 1$. In each case, dotted-green, dot-dashed black, dashed-red, and solid blue lines represent $\kappa_1 = \kappa_2 = \kappa = 0$, 0.20, 0.35, and 0.60, respectively.*

4.5 Gaussian Quantum Steering and Entanglement in Squeezed States

In this section, we explore the Gaussian quantum steering and entanglement when a special type of Gaussian states, such as squeezed states, are employed as input states.

In the lossless regime, squeezing generates a strong Gaussian steering and entanglement. The strong quadrature correlations which squeezed states possess lead to the high peak values of both quantum correlations. In addition, similar to the above cases where hierarchy relation was satisfied, here we see that for squeezed states the magnitude of quantum steering is less than that of quantum entanglement, following a strict hierarchy.

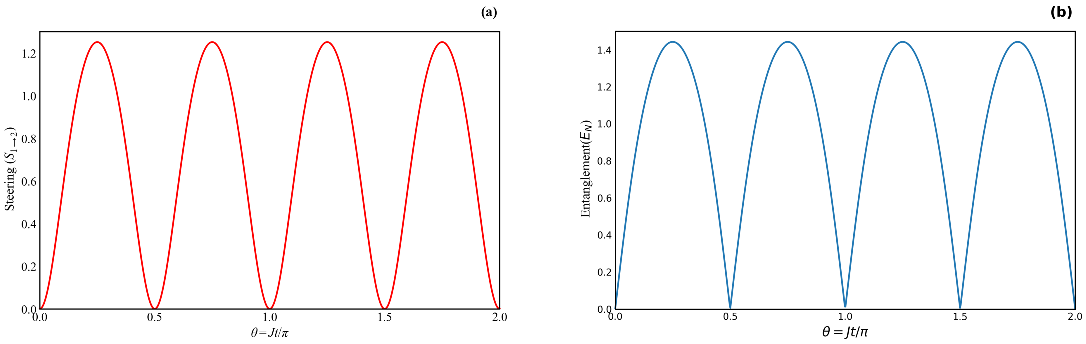

Figure 4.5: *Two kinds of quantum correlations: (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time, for fixed parameters* $\kappa_1 = \kappa_2 = \kappa = 0$ *and squeezing factor* $r = 0.5$.

To get further insight, we investigate the effect of loss parameter k on Gaussian quantum steering and entanglement when a special type of Gaussian squeezed states are used at the input, for example, see Fig. 4.6. From the figure, we see that loss parameter k degrades the strength as well as the time evolution of both quantum steering and entanglement and the degradation depends on the magnitude of loss rate k. Moreover, we observe that quantum steering is always less than entanglement in magnitude for every value of κ, which shows the hierarchy.

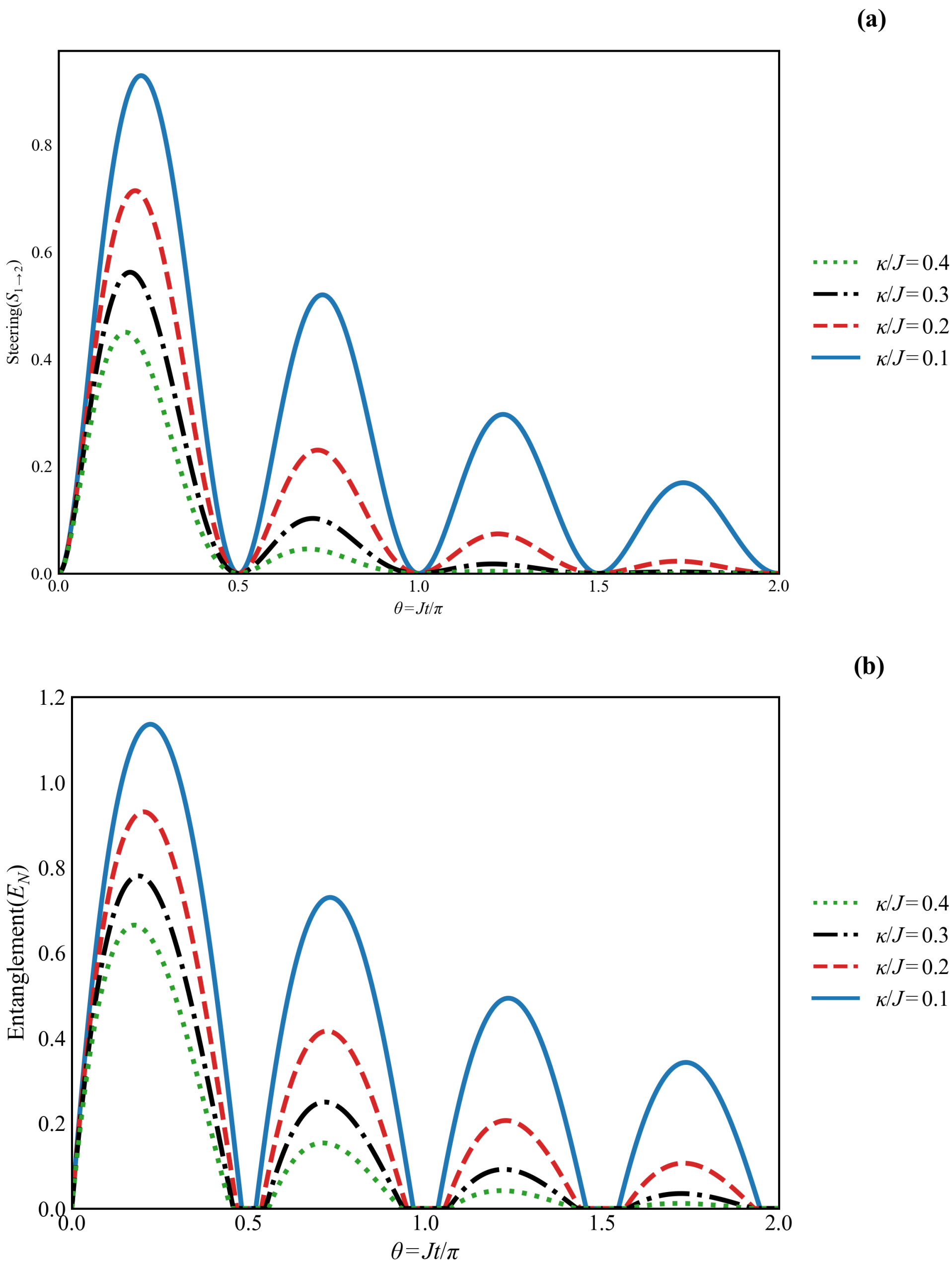

Figure 4.6: *Two kinds of quantum correlations: (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time, with squeezing factor* $r = 0.5$. *In each case, the solid blue, dashed-red, dot-dashed black, and dotted-green lines correspond to* $\kappa_1 = \kappa_2 = \kappa = 0.1$, 0.2, 0.3, *and* 0.4, *respectively.*

4.6 Effect of coupling strength on Quantum Steering and Entanglement

Finally, in this section we investigate the effect of the coupling strength J on Gaussian quantum steering and entanglement. The quantum correlations are plotted versus the scaled parameter κt, which represents the propagation time in the presence of loss. We see that, as the coupling becomes stronger, the strength and time evolution of both quantum steering and entanglement increases. This demonstrates that a higher coupling partially compensates the adverse impact of losses. Moreover, when the coupling parameter is stronger, the number of oscillations increases, which means that there is a high rate of coherent exchange of energy between the coupled modes. This indicates that better coupling enhances the generation of quantum correlations. The coupling parameter facilitates easier movement of photons between the waveguides hence increases the strength of quantum correlations between modes despite some level of loss is present. Conversely, when the coupling parameter is weaker the quantum correlations decays faster as they propagate. Moreover, quantum entanglement is always greater than quantum steering for all values of J, which proves the hierarchy between them.

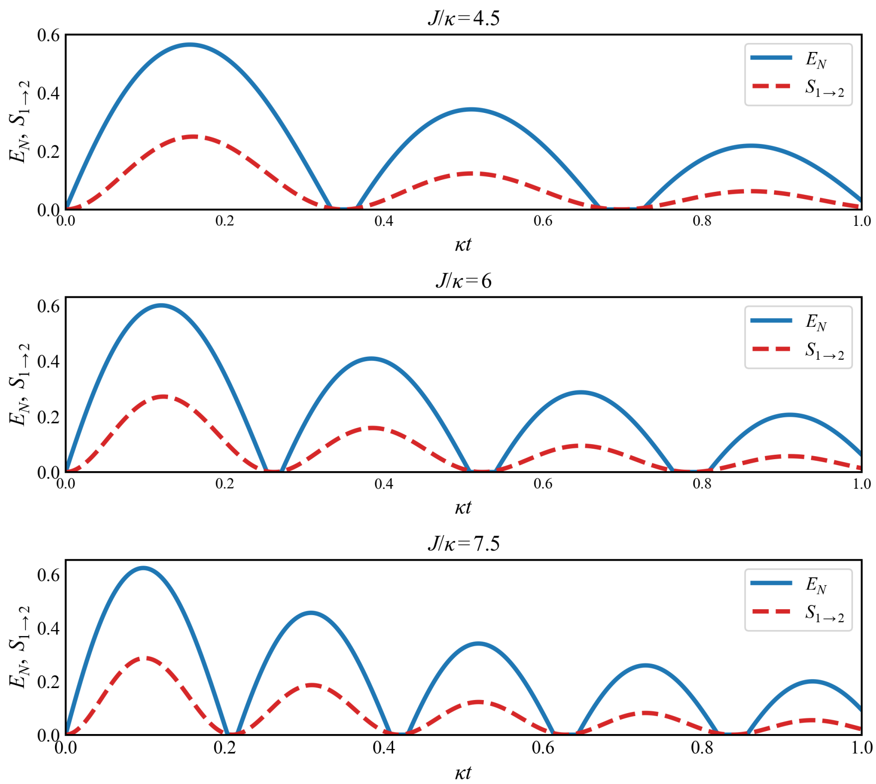

Figure 4.7: *Two kinds of quantum correlations: (a) quantum steering and (b) quantum entanglement are plotted against dimensionless interaction time κt for different coupling strengths J/κ=4.5, 6, and 7.5. In each panel, the solid-blue curve represents the entanglement E_N, while the dashed-red curve represents the Gaussian steering $S_{1\to 2}$.*

4.7 Summary

The key findings of this chapter are as follows: This chapter includes an in-depth exploration of the dynamical performance of Gaussian quantum entanglement and Gaussian quantum steering within a coupled lossy waveguide system using the covariance matrix framework. This analysis is done through the systematic investigation of the role of important physical parameters that determine the dynamics of the system. First of all we observe the overall time evolution of entanglement and steering by setting the nonclassicality, purity and loss parameters fixed. From our results, we find an oscillatory behavior of both quantum correlations as functions of the normalized propagation time, which is due to the coherent mode-coupling interaction between the waveguides. We see that maxima and minima in entanglement as well as steering take identical propagation lengths which validates their physical origins. As the process continues, quantum steering is always smaller in magnitude than quantum entanglement, which follows the hierarchy strictly. After that, the influence of nonclassicality is analyzed. It is revealed that the increase in nonclassicality positively affect the Gaussian quantum entanglement and steering. The effect of purity by the state is then examined. The decrease in purity results in degradation of the quantum entanglement and steering, thus showing the susceptibility of the two to noise and decoherence. Further, the effects of the loss is also examined. An increment in the loss parameter leads to a sharp reduction of the quantum correlations i,e., quantum entanglement and quantum steering. We then study the role of squeezing on quantum entanglement and steering. Squeezing in the absence of losses produces high quantum entanglement and steering due to the increase quadrature correlations. Both of the correlations decreases by introducing losses. Although loss decreases quantum correlations but it does not vanish them, they are correlated even with the effect of losses as well, which shows their robustness. Lastly, the influence of the strength of coupling is investigated. We see that high coupling increases evanescent coupling signifying more efficient transfer of photons, hence increases number of oscillation which produce great improvement in the amplitudes and persistence of entanglement and steering, which even compensate the effect of loss parameter. However, steering is always less than entanglement in magnitude which proves the hierarchy. On the whole, this chapter confirms that the nonclassicality, purity, loss, squeezing, and the strength of the coupling in coupled waveguide systems have a strong impact on the Gaussian quantum entanglement and steering. Both, quantum entanglement and steering, are found to be robust and steering is less than entanglement in magnitude under same physical state, which proves the hierarchy.

Chapter 5

Conclusion

In this thesis, the dynamical behaviour of Gaussian quantum steering and entanglement has investigated in a coupled lossy waveguide system using the covariance matrix formalism within the framework of continuous variable quantum optics. The reason behind the study was that, it is important to get insight on the impact of realistic dissipation and system parameters on the persistence and utility of quantum correlations in integrated photonic platforms [11, 22, 64].

The analysis has revealed that quantum steering and entanglement have oscillatory behaviour with respect to propagation time which is due to the coherent transfer of energy between the coupled waveguide modes. These oscillations are the evidence of the underlying evanescent mechanism of coupling and are a conclusive evidence of interaction between coherent modes within the system. However, the loss of correlations between the two modes in the presence of optical loss is what causes a gradual decay of both correlations, which in turn demonstrates the inevitable effect of dissipation in realistic photonic devices.

One of the key findings of the given work is that the hierarchical character of quantum correlations indeed exists, quantum steering is seen to be more vulnerable than quantum entanglement under the same physical conditions and is less in magnitude than entanglement. This behaviour is seen in all regimes of parameters studied and in total accord with the hierarchy between quantum entanglement and quantum steering [38, 65].

Furthermore, a systematic study of the quantum correlation with significant physical parameters has carried out. On one hand, the increase of the nonclassicality and purity is discovered to increase entanglement and steering, and thus increase their resistance to loss. On the other hand, an increased loss coefficient degraded the quantum correlations. The coupling parameter increased the rate of exchange of quantum correlations between the modes i.e., number of oscillation increased, which compensated the effect of loss parameter on quantum correlations. The study further indicated that as the ideal lossless systems can sustain strong quantum correlations, the realistic losses, although decreased the magnitude of quantum correlations, but entanglement and steering were still robust even in the presence of losses. These results are directly relevant to current challenges in quantum information processing, particularly in photonic quantum computing architectures where decoherence and loss severely limit scalability [11, 22, 64]. By explicitly modeling dissipation and quantifying the survivability of quantum correlations during propagation, this study provides a realistic assessment of how entanglement and steering behave under practical conditions. In this sense, the work did not eliminate fundamental limitations such as losses, but instead clarified the parameter regimes in which useful quantum correlations can still be preserved. To sum up, this research work has proved that useful amounts of Gaussian quantum steering and entanglement can be maintained in integrated waveguide systems.

Chapter 6

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